

VOLUME 2

SOLUTION 0-2

Cicada 3301 - Liber Primus pages 0-2

27 x 27 rune matrix / continuation beyond COLD



totient inheritance / hidden keys / Mobius phase / I MAY CRY

SECTION 01

Continuing beyond COLD

Volume 1 ends at AS I GO, THE WEATHER TURNS COLD. Volume 2 continues from the final state of COLD.

AS I GO, THE WEATHER TURNS COLD.

STRONGEST CONTINUATION

AS I GO, THE WEATHER TURNS COLD.
I MAY CRY...

The main new observation is that totient values appear to persist across stages of the route. In some cases, values already derived from a key or node are reused as movement distances and lead to the location of another key.

The exact rule that decides when this mechanism is activated is still unknown. However, the mechanism is visible in multiple strong examples.

STATUS

Proposed cryptanalytic research. Not an official Cicada 3301 solution.
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SECTION 02

Core rules used in Volume 2

The same small set of operations from Volume 1 remains the backbone of the continuation.

Mirrored 3-rune node compilation

```
a-b-a -> a-φ(b)-a
Example: H-TH-H -> H-U-H
because φ(TH=2)=1=U
```

Totient signature

```
signature(K) = (φ(k1), φ(k2), φ(k3))
```

Mobius phase

```
p = Σ μ(φ(ki)) mod 3
```

Gematria subtraction

```
plaintext = ciphertext - key mod 29
```

The phase is determined numerically before judging the plaintext. Gematria Primus values are the same 0-based values used in Volume 1.

SECTION 03

Two observed types of keys

The continuation appears to use both directly generated keys and keys that must be found elsewhere in the 27 x 27 grid.

TYPE A - DIRECTLY GENERATED

A mirrored 3-rune node directly generates a key by transforming its center with ϕ .

Example: H-TH-H -> H-U-H

TYPE B - HIDDEN KEYS

Specific totient values derived earlier in the route can be reused as movement distances. Those movements lead to the location of another key in the 27 x 27 grid.

Observed pattern: earlier totient values -> movement -> hidden key location

A second distinction appears in the current model. If a discovered key is non-mirrored, it is used directly: H-NG-C -> H-NG-C. If a discovered structure is mirrored, its center is transformed by ϕ first: NG-B-NG -> NG-T-NG.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27								
1	S/Z	H	E	O	G	M	IAIO	F	1	S/Z	Y	E	NG	C/K	TH	J	G	AE	1	F	J	OE	1	O	N	TH	1	N	E	W	B				
2	A	S/Z	OE	EO	I	NG	UV	1	N	AE	G	I	TH	1	H	B	N	IAIO	P	B	N	1	EO	J	TH	1	A	NG	Y	X	1	B			
3	D	P	1	E	IAIO	F	X	G	F	1	S/Z	P	T	F	P	UV	NG	Y	NG	X	IAIO	Y	H	1	B	UV	1	G	O	P	1	NG			
4	AE	TH	1	I	AE	H	X	TH	P	G	1	Y	F	NG	C/K	J	1	TH	A	M	I	NG	I	IAIO	Y	1	H	EO	1	T	H	AE			
5	IAIO	1	X	Y	1	E	UV	NG	1	C/K	F	N	1	EO	TH	J	I	TH	TH	P	1	Y	NG	EA	1	Y	N	E	Y	1	D	X	NG	W	1
6	B	C/K	X	D	B	F	1	M	T	N	E	1	EA	J	N	L	G	B	1	X	G	TH	4	Y	I	D	A	1	NG	G	M	J			
7	R	O	L	EO	1	Y	T	P	I	TH	D	J	1	OE	H	L	1	EA	1	F	X	AE	P	G	X	1	F	L	W	EO	T	AE	1		
8	F	G	N	1	M	EO	L	N	1	NG	M	F	R	IAIO	1	A	S/Z	1	EA	M	NG	X	X	EO	D	B	1	OE	D	M	O	1	EA		
9	EO	H	1	D	NG	G	1	OE	NG	L	UV	R	N	T	1	AE	SZ	Y	UV	H	R	T	1	H	C/K	1	SZ	OE	T	OE	NG	UV			
10	R	1	S/Z	1	F	O	E	W	EA	OE	L	F	AE	1	M	R	NG	1	D	A	M	R	1	W	A	I	M	SZ	N	1	SZ	OE	R		
11	UV	W	SZ	L	IAIO	OE	A	AE	NG	SZ	D	P	T	1	C/K	R	TH	IAIO	1	R	B	O	D	TH	F	1	Y	X	I	T	4				
12	OE	I	1	F	L	O	1	F	EA	1	R	O	OE	M	H	M	G	P	H	1	AE	H	N	O	H	Y	1	OE	SZ	L	1	N	G	1	
13	AE	Y	1	OE	C/K	B	L	J	1	NG	L	OE	AE	J	EA	W	1	EA	TH	O	1	B	OE	Y	1	C/K	F	C/K	J	1	L	AE	4	N	1
14	TH	P	1	UV	X	1	OE	X	G	1	P	F	SZ	EO	AE	OE	NG	P	EO	O	E	A	1	G	AE	IAIO	NG	TH	O	1	P	A			
15	OE	TH	R	NG	AE	1	C/K	SZ	B	EO	Y	H	1	T	X	E	L	R	4	G	W	Y	1	J	F	I	N	IAIO	C/K	Y	EA				
16	I	IAIO	1	O	TH	1	E	C/K	AE	C/K	X	IAIO	EA	1	O	L	C/K	1	F	R	D	NG	E	UV	1	D	C/K	L	EA	SZ	X	C/K			
17	G	IAIO	4	W	SZ	TH	1	F	D	NG	1	I	IAIO	M	A	AE	G	1	W	SZ	1	N	D	1	C/K	P	TH	X	P	EA	EA	1	W	UV	1
18	EA	W	1	F	W	J	Y	1	X	D	W	C/K	G	C/K	OE	1	D	X	OE	1	R	IAIO	G	1	N	P	A	Y	P	1	C/K	Y	H	1	
19	F	E	J	EA	N	1	OE	AE	1	UV	A	1	H	R	1	E	EA	TH	F	J	A	1	L	X	SZ	T	1	M	F	L	F	T	1	G	
20	I	TH	1	UV	L	G	1	X	EA	T	SZ	L	J	P	1	L	X	I	1	OE	M	1	UV	4	H	T	1	B	EO	L	D	H	B	M	1
21	EA	R	B	1	N	H	B	E	G	EO	1	D	L	W	EO	IAIO	P	O	1	H	E	EA	1	W	I	R	I	H	1	UV	TH	H			
22	Y	1	N	X	B	G	J	P	UV	1	NG	EA	F	N	I	E	D	IAIO	NG	R	1	D	B	J	IAIO	OE	M	I	F	T	1				
23	J	D	I	TH	R	L	SZ	1	E	EO	O	G	B	T	D	1	TH	A	N	C/K	Y	1	IAIO	SZ	TH	D	1	NG	F	L	E				
24	G	H	C/K	1	E	O	I	T	N	X	1	P	T	F	H	R	D	E	F	T	1	AE	W	H	1	B	C/K	1	F	1	P	A	L		
25	UV	EA	N	L	J	1	J	C/K	L	W	EA	D	UV	D	EO	1	EA	X	D	W	H	EA	1	TH	IAIO	AE	C/K	L	T	W					
26	E	I	C/K	1	P	OE	D	C/K	1	N	H	A	1	R	I	G	TH	EA	E	T	G	1	TH	H	O	IAIO	W	AE	J	E	1	NG			
27	EA	D	1	O	AE	1	A	L	1	B	J	C/K	UV	X	T	A	B	J	P	1	F	EA	1	H	D	N	IAIO	UV	P	SZ	UV	W	13		

The same 27 x 27 map, with the relevant Volume 2 structures highlighted.

SECTION 04

Frozen state at the end of COLD

Volume 1 derives the COLD key from H-TH-H $\rightarrow$ H-U-H because $\phi(\text{TH}=2)=1=U$. The ciphertext for COLD ends at A(11,7).

parent structure: H-TH-H
active/generated key: H-U-H
inherited totient signal: 4-1-4
COLD endpoint: A(11,7)

A structural fingerprint at the endpoint

The final A of COLD is immediately inside EA(10,7) - A(11,7) - EA(12,7), giving the mirrored structure EA-A-EA.

$\phi(\text{EA}=28)=12$
 $\phi(\text{A}=24)=8$
 $\phi(\text{EA}=28)=12$
EA-A-EA $\rightarrow$ 12-8-12

WHY THIS MATTERS

The later I MAY key has the exact same totient signature: 12-8-12.

SECTION 05

The inherited 4-1-4 signal locates B

Starting from the final A of COLD, reuse the inherited values from H-U-H -> 4-1-4.

```
A(11,7)
D4 -> S(15,7)
R1 -> B(15,8)
A(11,7) -> D4 -> S(15,7) -> R1 -> B(15,8)
```

I	EO	L	N	1	NG	M	F
G	G	1	OE	NG	L	U/V	R
	E	W	EA	OE	L	F	
	IA/IO	4	OE	A	AE	NG	S/Z
	O	1	F	EA	1	R	O
K	B	L	J	1	NG	L	OE'
	1	OE	X	G	1	P	F
	AE	1	C/K	S/Z	B	EO	Y
I	1	E	C/K	AE	C/K	X	IA/IC
Z	TH	1	F	D	NG	1	I

The local route from the COLD endpoint toward B, with the inherited distances applied.

GEOMETRIC TRANSITION

The values are not invented at B. They are inherited from the already generated COLD key signature 4-1-4.

SECTION 06

B reveals NG-B-NG

B is the center of NG(13,8) - B(15,8) - NG(17,8). The discovered structure is therefore NG-B-NG.

IO	OE	A	AE	NG
	F	EA	R	O
	L	J	NG	L
E	X	G	P	F
E	C/K	S/Z	B	EO
	C/K	AE	C/K	X
H	F	D	NG	I

B sits at the center of the mirrored NG-B-NG structure.

Because the structure is mirrored, transform the center:

$\phi(B=17)=16=T$
NG-B-NG -> NG-T-NG

The actual key is NG-T-NG. Its totient signature is:

$\phi(NG=21)=12$
 $\phi(T=16)=8$
 $\phi(NG=21)=12$
NG-T-NG -> 12-8-12

EXACT STRUCTURAL MATCH

EA-A-EA -> 12-8-12 <- NG-T-NG

The endpoint of COLD already contains the numerical fingerprint of the key later used for I MAY.

SECTION 07

Mobius phase fixes I MAY

For NG-T-NG -> 12-8-12, all three Mobius values are zero.

$\mu(12)=0$
 $\mu(8)=0$
 $\mu(12)=0$
 $p = 0$

Phase	Result
0	I M A Y
1	S X A TH
2	I X F Y

The newly found key is applied back to the preserved parent structure H-TH-H. Its center is TH(26,19). Reading left gives TH(26,19) - G(26,18) - T(26,17) - E(26,16).

Ciphertext: TH-G-T-E
Key: NG-T-NG-NG
TH(2) - NG(21) mod 29 = 10 = I
G(6) - T(16) mod 29 = 19 = M
T(16) - NG(21) mod 29 = 24 = A
E(18) - NG(21) mod 29 = 26 = Y

I MAY

SECTION 08

The I MAY chain

The whole transition can be written as one inherited chain from the old COLD parent structure to the new plaintext.

H-TH-H
| $\phi(\text{TH})=\text{U}$
v
H-U-H
| totient signature
v
4-1-4
| after COLD: D4, R1
v
B(15,8)
v
NG-B-NG
| $\phi(\text{B})=\text{T}$
v
NG-T-NG
| signature 12-8-12
| Mobius phase 0
v
return to H-TH-H
| read TH-G-T-E
v
I MAY

D	1	TH	A	N	C/K	Y	1	IA
R	D	E	F	T	1	AE	V	
EO	1	EA	X	D	W	H	E	
TH	EA	E	T	G	1	TH	I	

The preserved H-TH-H structure and the leftward I MAY ciphertext read.

CROSS-CHECK

EA-A-EA -> 12-8-12 <- NG-T-NG

SECTION 09

The final E opens the next node

The ciphertext for I MAY ends at E(26,16). That same rune is immediately part of a new mirrored structure.

E(24,16) - X(25,16) - E(26,16)
E-X-E

X	E	L	R 4	G	V
L	C/K 1	F	R	D	N
AE	G 1	W	S/Z 1	N	I
OE 1	D	X	OE 1	R	IA/
EA	TH	F	J	A 1	I
L	X	I	OE	M 1	U/
EO	IA/IO	P	O 1	H	E
I	E	D	IA/IO	NG	F
T	D 1	TH	A	N	C/
H	R	D	E	F	T
D	EO 1	EA	X	D	V
G	TH	EA	E	T	C

The final E of I MAY is also the lower endpoint of E-X-E.

This repeats an important behavior of the model: the end of one read can become part of the next structural node.

CONTINUITY

The next node is not searched for independently. It is attached directly to the terminal rune of the preceding ciphertext.

SECTION 10

E-X-E generates E-G-E

Because E-X-E is mirrored, its center is transformed with ϕ .

```
 $\phi(X=14)=6=G$   
E-X-E -> E-G-E
```

The new key has the totient signature:

```
 $\phi(E=18)=6$   
 $\phi(G=6)=2$   
 $\phi(E=18)=6$   
E-G-E -> 6-2-6
```

Mobius phase:

```
 $\mu(6)=+1$   
 $\mu(2)=-1$   
 $\mu(6)=+1$   
 $p = (+1 - 1 + 1) \bmod 3 = 1$   
active key = G-E-E
```

Phase	Result
0	OE T Y
1	C R Y
2	OE R N

Again, the Mobius rule selects the phase numerically before the readable output is preferred.

SECTION 11

The inherited value 6 leads to J

Using the outer totient value 6, move upward from the center X.

X(25,16) -> U6 -> J(19,16)
J is the center of OE(18,16) - J(19,16) - OE(20,16)
Reading upward from J: J(19,16) - OE(18,16) - S(17,16)

Decrypting CRY

Ciphertext: J-OE-S
Active key: G-E-E
 $J(11) - G(6) = 5 = C$
 $OE(22) - E(18) = 4 = R$
 $S(15) - E(18) \bmod 29 = 26 = Y$
J-OE-S - G-E-E = C-R-Y

CRY

RESULT

AS I GO, THE WEATHER TURNS COLD. I MAY CRY...

SECTION 12

The CRY chain

The second continuation follows the same recurring pattern: terminal rune -> mirror -> ϕ (center) -> signature -> phase -> inherited movement -> next read.

```
I MAY
| final ciphertext rune E
v
E-X-E
|  $\phi(X)=G$ 
v
E-G-E
| totient signature
v
6-2-6
| Mobius phase 1
v
G-E-E
| U6 from X
v
J(19,16)
| read J-0E-S upward
v
CRY
```

The strongest feature is not simply that CRY is readable. It is that the key phase, the movement value, and the target center are all already constrained by the preceding structure.

SECTION 13

Totient values do not always disappear

The strongest broader observation of Volume 2 is that a totient result may remain relevant after its first use.

A value can appear first as part of a key signature and later reappear as a distance or structural relation elsewhere in the grid.

In the current model, totients can participate in several roles:

1. transform the center of a mirrored node
2. define a key signature
3. determine Mobius phase
4. act as movement distances
5. lead to hidden key locations

TOTIENT INHERITANCE

This behavior is referred to here as a totient chain. It does not mean that every earlier totient value is always reused. The exact activation rule is still unknown. What is observable is that specific derived values recur at structurally important transitions.

SECTION 14

H-NG-C and NG-B-NG

The current model distinguishes between a non-mirrored hidden key and a mirrored hidden node.

H-NG-C - NON-MIRRORED

Previously derived values are reused as distances in the grid: one value locates NG, another defines the vertical distance to H and C. The resulting key H-NG-C is non-mirrored and is therefore used directly.

NG-B-NG - MIRRORED

After COLD, H-U-H -> 4-1-4. Those values lead to B: A -> D4 -> R1 -> B. B is the center of NG-B-NG. Because the structure is mirrored, $\phi(B)=T$, so NG-B-NG -> NG-T-NG.

Y	H	1	B	U/V	1	G
I	IA/IO	Y	1	H	EO	1
Y	N	E	Y	1	D	
Y	I	D	A	1	NG	
G	X	1	F	L	W	
EO	D	B	1	OE	D	
H	C/K	1	S/Z	OE	T	
A	I	M	S/Z	N	1	
O	D	TH	F	1	Y	
O	H	Y	1	OE	S/Z	
Y	1	C/K	F	C/K	J	1
G	AE	IA/IO	NG	TH		
J	F	I	N	IA/IO		
U/V	1	D	C/K	L	EA	
P	TH	X	P	EA		
N	P	A	Y	P	1	
S/Z	T	1	M	F	L	
T	1	B	EO	L	D	
W	I	R	I	H	1	
B	J	IA/IO	OE	M		
IA/IO	S/Z	TH	D	1	NG	
W	H	1	B	C/K	1	F

A vertical overview of the relevant hidden-key geometry in the shared 27 x 27 map.

The new NG-T-NG key is then applied back to the preserved H-TH-H structure and produces I MAY.

SECTION 15

Current rule summary

The working model can be summarized as two linked mechanisms.

```

mirrored 3-rune node
|
v
apply  $\phi$  to center
|
v
key
|
+--> totient signature
| |
| +--> Mobius phase
| |
| +--> possible inherited movement values
|
v
decrypt ciphertext
|
v
next structural location

```

```

earlier derived totient values
|
v
movement through 27 x 27 grid
|
v
hidden key / hidden node
|
+--> mirrored: transform center with  $\phi$ 
+--> non-mirrored: use directly

```

SECTION 16

What is supported - and what is still open

The arithmetic and geometry below are reproducible. The complete state machine is not yet fully formalized.

Strongly supported

H-U-H -> 4-1-4 is inherited from the COLD stage.
D4 -> R1 reaches B at the center of NG-B-NG.
NG-B-NG -> NG-T-NG follows the same center-totient rule.
EA-A-EA and NG-T-NG share the exact signature 12-8-12.
Mobius fixes NG-T-NG to phase 0 before plaintext inspection.
TH-G-T-E decrypts exactly to I MAY.
The final E of I MAY immediately opens E-X-E.
E-X-E -> E-G-E follows the same mirrored-center rule.
Mobius fixes E-G-E to phase 1 before plaintext inspection.
U6 lands exactly on J, another mirrored center.
J-0E-S decrypts exactly to CRY.

Open questions

What exact condition activates hidden-key search? Why are particular inherited totient values used at a given transition? What fully deterministic rule selects movement direction? How is parent-state return represented formally? Can the same rules predict the continuation after CRY without adding a new rule?

SECTION 17

The strongest result of Volume 2

After repeated testing and attempts to break the structure, the strongest continuation found so far remains:

AS I GO, THE WEATHER TURNS COLD.
I MAY CRY...

I MAY is currently the strongest post-COLD result. CRY is the strongest continuation found after I MAY. Later candidate branches remain experimental and are not included as fixed plaintext in this volume.

Research conclusion

Volume 2 suggests that Euler totients are doing more than transforming individual rune nodes. They appear to connect mirror symmetry, key generation, numerical signatures, Mobius phases, grid movement, hidden-key locations, and structural inheritance between stages.

The system appears strongly centered on mirrored geometry: when a mirrored structure is encountered, the center repeatedly becomes the point where the totient transformation acts.

MAIN HYPOTHESIS

The most important new result is not only the plaintext candidate I MAY CRY, but the possibility that totient values form a persistent control layer governing both cryptographic and spatial behavior in the 27 x 27 grid.

APPENDIX

Minimal machine-readable chain

Part 1 - COLD to I MAY

```
VOLUME_1_END:
plaintext = "AS I GO, THE WEATHER TURNS COLD"
endpoint = A(11,7)
parent_node = H-TH-H
cold_key = H-U-H
cold_signature = 4-1-4
```

```
POST_COLD:
A(11,7)
D4 -> S(15,7)
R1 -> B(15,8)
```

```
NODE:
NG(13,8)-B(15,8)-NG(17,8)
```

```
COMPILE:
 $\phi(B=17)=16=T$ 
NG-B-NG -> NG-T-NG
```

```
SIGNATURE:
NG-T-NG -> 12-8-12
```

```
CROSS_CHECK:
EA-A-EA -> 12-8-12
```

```
PHASE:
 $\mu(12)+\mu(8)+\mu(12)=0$ 
phase=0
```

```
CIPHERTEXT:
TH-G-T-E
```

```
KEY:
NG-T-NG-NG
```

```
PLAINTEXT:
I-M-A-Y
```

APPENDIX

Minimal machine-readable chain

Part 2 - I MAY to CRY

```
NEXT_NODE:
E-X-E

COMPILE:
 $\phi(X=14)=6=G$ 
E-X-E -> E-G-E

SIGNATURE:
E-G-E -> 6-2-6

PHASE:
 $\mu(6)+\mu(2)+\mu(6)=1$ 
phase=1
active_key=G-E-E

MOVE:
X(25,16)
U6 -> J(19,16)

CIPHERTEXT:
J-OE-S

PLAINTEXT:
C-R-Y

CURRENT_RESULT:
"AS I GO, THE WEATHER TURNS COLD. I MAY CRY..."
```

I MAY CRY...